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Introduction to Pandas and Series Data
DataFrame

Week 2 - Assignment

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- Programming Assignment: Assignment 2 3h

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1. For the following code, which of the following statements will not return True?

1 / 1 point

Try again

```
1 import pandas as pd
2 sdata = {'Ohio': 35000, 'Texas': 71000, 'Oregon': 16000, 'Utah': 5000}
3 obj1 = pd.Series(sdata)
4 states = ['California', 'Ohio', 'Oregon', 'Texas']
5 obj2 = pd.Series(sdata, states)
6 obj3 = pd.isnull(obj2)
```

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☐

```
1 x = obj2['California']
2 obj2['California'] != x
```

☒

```
1 obj2['California'] == None
```

☐

```
1 obj3['California']
```

☐

```
1 import math
2 math.isnan(obj2['California'])
```

☒ Correct
The value of obj2['California'] is nan which is not the same as None, so this will return False

2.

```
1 import pandas as pd
2 d = {'1': 'Alice', '2': 'Bob', '3': 'Rita', '4': 'Molly', '5': 'Ryan'}
3 s = pd.Series(d)
```

1 / 1 point

In the above python code, the keys of the dictionary **d** represent student ranks and the value for each key is a student name. Which of the following can be used to extract rows with student ranks that are lower than or equal to 3?

☐ S.iloc[0:2]

☐ S.loc[0:2]

☒ S.iloc[0:3]

☐ S.loc[0:3]

☒ Correct
S.iloc[i:j] can be used to retrieve Series rows from indices i to j-1

3. Suppose we have a DataFrame named **df**. We want to change the original DataFrame **df** in a way that all the column names are cast to upper case. Which of the following expressions is **incorrect** to perform the same?

1 / 1 point

☐ df = df.rename(mapper = lambda x: x.upper(), axis = 'columns')

☐ df.rename(mapper = lambda x: x.upper(), axis = 1, inplace = True)

☐ df = df.rename(mapper = lambda x: x.upper(), axis = 1)

☒ df.rename(mapper = lambda x: x.upper(), axis = 1)

☒ Correct
This is incorrect because the rename method will return a new DataFrame by default. We have to pass the result to our original DataFrame **df** or set the inplace parameter to 'True'.

4.

1 / 1 point

	gre score	toefl score
Serial No.		
1	337	118
2	324	107
3	316	104
4	322	110
5	314	103

For the given DataFrame **df** we want to keep only the records with a **toefl score** greater than 105. Which of the following will **not** work?

☐ df.where(df['toefl score'] > 105).dropna()

☒ df.where(df['toefl score'] > 105)

☐ All of these will work

☐ df[df['toefl score'] > 105]

☒ Correct
This will not work as **df.where()** will not drop any data we don't want, it will just set their values to **nan**.

5. Which of the following can be used to create a DataFrame in Pandas?

1 / 1 point

☐ 2D ndarray

☐ Pandas Series object

☐ Python dict

☒ All of these work

☒ Correct
All of these can be used to create a DataFrame in Pandas

6. Which of the following is an **incorrect** way to **drop** entries from the Pandas DataFrame named **df** shown below?

1 / 1 point

	one	two	three	four
Ohio	0	1	2	3
Colorado	4	5	6	7
Utah	8	9	10	11
New York	12	13	14	15

☐ df.drop('Ohio')

☐ df.drop(['Utah', 'Colorado'])

☐ df.drop('one', axis = 1)

☒ df.drop('two')

☒ Correct
This is an incorrect way to drop values from the column named 'two' because the axis has not been specified as 1 (representing 'columns') and the default value of axis is 0. It would yield the following error: **KeyError: ['two'] not found in axis.**

7. For the Series **s1** and **s2** defined below, which of the following statements **will give an error**?

1 / 1 point

```
1 import pandas as pd
2 s1 = pd.Series([1: 'Alice', 2: 'Jack', 3: 'Molly'])
3 s2 = pd.Series({'Alice': 1, 'Jack': 2, 'Molly': 3})
```